

# Chapter 1

## Conclusion

This final chapter reviews the work presented in this thesis. It begins by revisiting each research question based on the contributions presented. It then discusses the limitations of these contributions. Next, it presents the directions for future research that these limitations open. Finally, it closes with a broader reflection on this work.

### 1.1 Summary of Contributions

This section revisits the four research questions introduced in Section ?? . For each of them, it summarizes the answer provided by this thesis.

Research Question ?? concerned the approaches proposed in the literature to analyze the internal workings of Neural Networks (NNs). To answer it, we first made explicit the working hypothesis shared by most of this literature, the *Concept Hypothesis* (Hypothesis ??). According to this hypothesis, NNs generalize by organizing their Latent Spaces (LSs) around concepts. We then organized existing methods into four families: probing, interpretable directions, Sparse Autoencoders (SAEs), and clustering.

Research Question ?? concerned the organization of existing Explainable Reinforcement Learning (XRL) methods [1, 2]. To answer this question, we first clarified what an explanation is. Building on this framework, we identified the concepts and sources of explanation. Regarding the subject of explanation, we distinguish two levels: a single decision and the agent’s strategy. For the source of explanation, we identify three possibilities: policies, trajectories, and value functions. From these elements, we proposed a taxonomy that accounts for the specificities of Reinforcement Learning (RL) and used it to review the state of the art.

Research Question ?? concerned the use of Latent Representations (LRs) to identify the mechanisms underlying the decisions of RL agents. To answer it, we introduced Clustering Agent Representation (CAR), an XRL method based on the principles of Mechanistic Explainability (ME) [3]. CAR trains several surrogate policies independently to reproduce the same agent. It then clusters their LRs and compares the resulting partitions. This procedure relies on the *Representation Convergence Hypothesis* (Hypothesis ??), according to which independently trained networks develop similar concepts for a given task.

Applied to six Atari environments, CAR recovers similar clustering structures across surrogate policies. The qualitative analysis shows that these clusters capture interpretable aspects of the task, such as episode progression, specific game situations, and behavioral patterns.

Research Question ?? concerned the objective quantification of explanation quality. To answer it, we introduced three metrics. The *Clustering Universality* (CU) measures the consistency of the clusters across independently trained surrogate policies. The *Abstraction Score* (AS) measures how far these clusters depart from the structure of the observations, of the actions, and of an untrained network. The experiments show that these two properties evolve in opposite directions across the network. The earliest layers are the most consistent, but also the least abstract. The *Adjusted Clustering Universality* (ACU) combines both properties into a single measure and provides a practical criterion for selecting the layer to analyze.

Taken together, these contributions form a coherent chain. The *Mechanistic Hypothesis* (Hypothesis ??) states that NNs can be explained through their internal mechanisms. The *Concept Hypothesis* (Hypothesis ??) states that these mechanisms are organized around concepts. The *Representation Convergence Hypothesis* (Hypothesis ??) states that these concepts reflect the task. CAR translates these hypotheses into an empirical procedure, and the experiments provide support for them.

## 1.2 Limitations

Although the research presented in this thesis has produced encouraging results, it remains subject to limitations. This section discusses the four main limitations identified at this stage.

First, CAR describes how the LS of an agent is organized, but not what each cluster means. The metrics quantify the consistency and abstraction of the clusters, but they say nothing about their content. In this thesis, the meaning of each cluster was assigned through manual inspection of the corresponding observations. This step is time-consuming, subjective, and difficult to scale to a large number of clusters or environments.

Second, the clusters identified by CAR are not yet directly related to the decisions of the agent. We observe that the agent distinguishes certain situations, but not how these distinctions lead to a given action. As a result, CAR cannot yet explain a specific failure or reveal a bias in the behavior of the agent.

Third, the experiments were conducted on six Atari environments. These environments are diverse, but they remain limited in size and complexity. Whether CAR scales to more complex environments and larger networks has not yet been established. In addition, only one clustering method was evaluated, based on  $k$ -means with Euclidean distance. Other distances and other clustering methods could reveal different structures in the LS. Since the pipeline of CAR is modular, these alternatives can be tested and compared using the proposed metrics.

Finally, the proposed metrics have only been applied to clustering. The other families of methods reviewed in Chapter ?? have not yet been evaluated within this framework. This is particularly the case for methods based on the *Superposition Hypothesis* (Hypothesis ??). These methods are widely used, but their assumptions remain debated. Our metrics could be used to test these

assumptions quantitatively, and to compare these methods with CAR on the same tasks.

## 1.3 Perspectives

The limitations discussed above open several directions for future research. This section presents four of them. Each direction is presented as a research topic in its own right. We first state the research question and the main scientific challenge it raises. We then describe the approach we envisage and how this approach could be validated.

### 1.3.1 Automatic Textual Description of Clusters

This direction aims to automatically obtain a reliable textual description of each cluster. The main challenge is that a cluster often captures a dynamic rather than a static situation. It must therefore be described from sequences of observations rather than from isolated images.

For each cluster, the video segments formed by consecutive observations belonging to it are first extracted. Each segment is then described in natural language by a vision-language model. Finally, the descriptions of all segments of a cluster are synthesized into a single description at the level of the cluster.

The reliability of these descriptions can be assessed through a classification test. Given a new video segment, not used to produce the descriptions, a model must identify the cluster to which it belongs, using only the textual descriptions of the clusters. The predicted cluster is then compared with the one assigned by CAR. If the classification accuracy is high, the textual description faithfully captures what distinguishes the cluster. This would fully automate the CAR method, from the identification of the clusters to their interpretation.

### 1.3.2 From Latent Similarity to the Strategy of the Agent

This direction aims to derive an explanation of the strategy of the agent from the similarity of its LRs. The main challenge is that the clusters only group states that the agent perceives as similar. They must be turned from a static view of the LS into a dynamic view of the behavior of the agent.

Each cluster can be treated as an abstract state. The transitions observed between clusters during episodes, together with the actions taken in each cluster, then define a simplified Markov Decision Process (MDP). This MDP can be represented as a high-level graph. Its nodes are the concepts identified by CAR, and its edges describe how the agent moves between them. This graph provides a summary of the interaction between the agent and its environment, and therefore of its strategy.

Such a graph is useful only if it is truly informative for an observer. An observer given the graph should anticipate or describe the behavior of the agent better than an observer without it. For example, the observer could be asked to predict the next action of the agent, or the next phase of the game. This validation relies on the protocols presented in the next direction.

### 1.3.3 Human Evaluation of Explanations

This direction aims to determine whether the explanations produced by CAR actually improve the observer’s mental model of the agent. It also aims to determine whether the metrics proposed in this thesis, namely CU, AS, and ACU, are good indicators of this improvement. The main challenge is that understanding has no consensual operational definition, which makes it difficult to measure.

Experimental protocols from disciplines such as cognitive science and psychology can be adapted to this setting. In a forward simulation task, the observer predicts the action of the agent in a given situation. In a counterfactual task, the observer predicts how the action would change if the situation were modified. In a failure detection task, the observer identifies the situations in which the agent behaves incorrectly.

A controlled experiment can compare several groups of observers. One group receives no explanation, one receives the explanations of CAR, and one receives those of an existing method, such as saliency maps. Statistical tests then determine whether the explanations significantly improve performance on these tasks. The gains observed can finally be compared with the values of CU, AS, and ACU. A strong correlation would confirm that these metrics are relevant indicators of the quality of an explanation.

### 1.3.4 Hypotheses on the Structure of the Latent Space

CAR assumes that concepts are organized as clusters, whereas the *Superposition Hypothesis* (Hypothesis ??) assumes that they are encoded as directions. This direction aims to determine which of these hypotheses best describes the LS of a RL agent. The main challenge is that the two families of methods do not produce the same kind of object, a partition in one case and a set of directions in the other, which makes them difficult to compare.

The metrics proposed in this thesis can serve as a common framework. They can be applied to the concepts extracted by methods based on superposition, provided that these concepts are expressed as a partition of the observations. The modularity of CAR also makes it possible to test other distances and other clustering methods within the same pipeline.

Each hypothesis can then be evaluated on the same agents and the same environments, using the same metrics. The hypothesis that yields the most universal and the most abstract concepts is the one that best describes the structure of the LS.

## 1.4 Final Remarks

To conclude, we can step back and consider this work as a whole. In XRL, a common approach is to replace NNs with simpler models that can be directly interpreted. The approach followed in this thesis is different. Rather than viewing the NN as an obstacle, we propose explainability methods that build on its specificities. From this perspective, explainability is not achieved despite the use of NNs, but through them. The results presented in this thesis show that this approach is feasible. It therefore offers a way to understand our creations and to ensure that they remain aligned with our intentions.